Mathematics Training Exercise (Numbers)

(Notes by Tat Wing Leung 22,29 Sept., 2018)

Divisibility

Some Divisibility Tests (*N* = 114043, in fact it is a prime number)

Divisors	Reduction	Reason
7,11,13	$114-043=71$	$7\times11\times13=1001$
37	$114+043=157$	$37\times27=999$
23, 29	$114-2(043)=28$	$23\times29\times3=2001$
31, 43	$114+4(043)=286$	$31\times43\times3=3999$
19	$1140+4(43)=1312$	$19\times21=399$
17, 47	$1140+8(43)=1484$	$17\times47=799$
41	$1140+16(43)=1828$	$41\times39=1599$
67	$1140-2(43)=1054$	$67\times3=201$
89	$1140-8(43)=796$	$89\times9=801$
53	$1140-9(43)=753$	$53\times17=901$

1. Find the GCD of 952 and 4301.
2. The number $\overline{65xy13}$ is divisible by 99, find all possible value(s) of $x+y$.
3. The six-digit number $\overline{a2006b}$ is divisible by 72, find *a* and *b*.
4. Find the unit digit of the number $1!+2!+3!+...+2009!$. (Note: $n!=n(n-1)(n-2)...1$.)
5. If the 9-digit number $\overline{123xyz789}$ is divisible by 999, find the value of $x+y+z$.
6. What is the remainder when $1^3+2^3+\cdots+2009^3$ is divided by 7?
7. Find the remainder when $1^3+6^3+11^3+16^3+\cdots+2001^3$ is divided by 2002.
8. $n+3$ divides $n^3+3n+2006$ and n is a 3-digit number. Determine the largest possible value of *n*.
9. Find the smallest positive integer *n* such that $999999n=111...11$.
10. Find the last two digits of $14^{14^{14}}$.
11. Does it exist a multiple of 2009 of the form 11…1?
12. Let *n* be an odd integer, show that the last two digits of $2^{2n}(2^{2n+1}-1)$ is 28.
13. Let $a_1,a_2,\cdots a_n$ be a permutation (rearrangement without omission nor repetition) of 1, 2, …, *n*. If *n* is odd, show that $(a_1-1)(a_2-2)\cdots(a_n-n)$ is even. Is the claim true if *n* is even?
14. If *p* is a prime number, show that $C_i^p=\dfrac{p!}{i!(p-i)!}, 1\le i\le p-1,$ is always divisible by *p*. (You may assume C_i^p is always a positive integer.) Is the claim true if *p* is not a prime number?

15. Show that (a) $10n+b$ is divisible by 7 if and only if $n+5b$ is, (b) $10n+b$ is divisible by 13 if and only if $n+4b$ is, (c) $10n+b$ is divisible by 17 if and only if $n+12b$ is.
16. Let n be an odd positive integer. Then the largest positive integer k, such that $n^{12}-n^8-n^4+1$ is divisible by 2^k for every n is A. 6 B. 7 C. 8 D. 9 E. 10?
17. How many positive integers n are there such that $8n+50$ is a multiple of $2n+1$?
18. Given two relatively prime natural numbers a and b, show that the greatest common factor of $a+b$ and a^2+b^2 is 1 or 2.
19. Find the least integer n with the following property: given n distinct integers $a_1, a_2, ..., a_n$, there are two whose sum or difference is divisible by 9.
20. Let p be a prime number. Show that there exists a prime number q such that for every integer n, the number $n^p - p$ is not divisible by q.
21. Let $n \geq 0$, define $F_n = 2^{2^n}+1$. (Fermat numbers) (a) If $m \neq n$, $d \mid F_n$, then d does not divide F_m. Hence show there exist infinitely many primes. (b) $F_{n+1} = F_n ... F_0 + 2$. (c) $A_1 = 2$, $A_{n+1} = A_n^2 - A_n$, $n \geq 1$. Let $n \neq m$. If $1 < d \mid A_n$, then d does not divide A_m. Hence show there exist infinitely many primes. (d) $A_{n+1} = A_n ... A_1 + 1$.
22. If there are finitely many primes $p_1, p_2, ..., p_s$, then for any integer N, $\sum_{n=1}^{N} \frac{1}{n} < (1-\frac{1}{p_1})^{-1} ... (1-\frac{1}{p_s})^{-1}$.
23. Let $N \geq 2$. (a) $\sum_{p<N} \frac{1}{p-1} > \ln\ln(N+1)$; (b) $\sum_{p<N} \frac{1}{p} > \ln\ln(N+1) - 1$.
24. (a) Any positive integer a is of the form $k^2 l$, where l is a product of distinct primes or 1; (b) $N \geq 2$, then $N \leq \sqrt{N} \cdot 2^{\pi(N)}$, where $\pi(x)$ is the number of primes less than or equal to x. (c) $\pi(N) \geq \frac{\log_2 N}{2}$; (d) The n^{th} prime $p_n \leq 2^{2n}$.
25. Let n be a positive integer with the following property: $2^n - 1$ divides a number of the form $m^2 + 81$, where m is a positive integer. Find all possible n.

26. Let k be a positive integer such that $p = 8k+5$ is a prime number. The integer $r_1, r_2, ..., r_{2k+1}$ are chosen so that the numbers $0, r_1^4, r_2^4, ..., r_{2k+1}^4$ give pairwise different remainders modulo p. Prove that the product $\prod_{1 \leq i < j \leq 2k+1} (r_i^4 + r_j^4)$ is congruent to $(-1)^{k)k+1)/2}$ modulo p.

27. Let $1 = d_0 < d_1 < ... < d_m = 4k$ be all positive divisors of $4k$, where k is a positive integer. Prove that there exists $i \in \{1, 2, ..., m\}$ such that $d_i - d_{i-1} = 2$.

Numbers

1. (a) Show that $1+\frac{1}{2}+...+\frac{1}{n}, n \geq 2,$ is never an integer. (b) Show that $1+\frac{1}{3}+...+\frac{1}{2n-1}, n \geq 2,$ is never an integer.

2. Determine all pairs (m,n) of positive integers for which 2^m+3^n is a square.
3. Prove that n^4+4^n is not a prime number for any integer $n>1$.
4.Find a 4-digit square number such that when each digit is increased by 3, the resulting 4-digit number is still a square.
5. Is $1\underbrace{00...0}_{2009\ 0's}1$ a prime?
6. If n is a positive integer and $n^2 = 11112222^2 + 3333^2 + 3334^2$, find n.
7. Show that for any natural number n, $\frac{21n+4}{14n+3}$ is irreducible (i.e. the fraction cannot be further reduced).
8. A positive integer n is *good* if the digits of n can be divided into two groups such that the sum of one group of the digits in one group is equal to the sum of the other, (for example, 1432 is good). Find the smallest possible n such that both n and $n+1$ are good.
9. Show that the product of four consecutive numbers is not a cube.
10. Find the least number whose last digit is 7 and which becomes 5 times larger when this last digit is carried to the beginning of the number.
11. There are five positive integers. Taken four at a time, they sum to four distinct values, namely 14, 15, 19 and 20. Find the sum of the five numbers.
12. A number n equals to the sum of squares of its four smallest divisors, find the largest prime factor of n.
13. For any m, there exist infinitely many n such that m^4+n is composite.
14. Can you find a natural number n such that $n!$ ends in exactly 2009 zeros?
15. Find all 4-digit numbers $\overline{aabb}$ which is a perfect square.
16. It is known that 999973 has *exactly* three distinct prime factors. Find the sum of these three prime factors.
17. Find the 4-digit number $\overline{abcd}$ such that $4\overline{abcd} = \overline{dcba}$.
18. If $2^8+2^{11}+2^n$ is a perfect square, find n.
19. Find the smallest multiple of 63 which contains only the digits 8 or 3.
20. Find the smallest multiple of 2834 which contains only the digits 0, 1 or 9.
21. What is the unit digit of $(2+1)(2^2+1)(2^4+1)(2^8+1)(2^{16}+1)(2^{32}+1)(2^{64}+1)$? (Use factorization of a^2-b^2.)
22. If x, y, z and n are natural numbers with $x^n+y^n=z^n$, then show that x, y and z are all greater than n.
23. If $\frac{1}{a}+\frac{1}{b}=\frac{1}{c}$ where a, b, c are positive integers with no common factor, show that $a+b$ is a square.
24. Start from Machine A, (m, n) becomes (n,m); while Machine B and C will produce $(m+3n,n)$ and $(m-2n,n)$ respectively. Start from (0, 1) and put into anyone of the

machines, after finitely many rounds, which of the following pairs cannot be produced?
A. (2009,1016) B. (2009,1004) C. (2009,1002) D. (2009,1008)
E. (2009,1032)

25. Let p be a prime of the form $3k+2$. Show that any number modulo p is a cube.

26. Show that for any integer $n>1$, $\frac{2^n-1}{n}$ is not an integer. (Use Fermat's little theorem and consider the smallest prime factor of n.

27. There exist infinitely many n such that n divides 2^n+2. Find one between 100 and 2000.

28. Find one pair of positive integers a, b such that (1) $ab(a+b)$ is not divisible by 7; (2) $(a+b)^7-a^7-b^7$ is divisible by 7^7.

29. (a) Show that 3^{n+1} just divides $2^{3^n}+1$ by induction.
(b) Find an integer n, where n is not purely a power of 3, and n divides 2^n+1.
(c) Is it possible to find n with exactly 2009 prime divisors, and such that n divides 2^n+1.

30. The real number $0.a_1a_2a_3...$ is defined as follows, a_n is the first non-zero digit (counted from the right) of $n!$. For instance, $1!=1$, hence $a_1=1$. Likewise $a_2=2$, $a_3=6$, $a_4=4$, $a_5=2$ since $5!=120$. Is the number rational?

31. Given positive integers a and b and such that $ab+1$ divides a^2+b^2, then $\frac{a^2+b^2}{ab+1}$ is a perfect square.

32. (a) If $n\left|2^n-2\right.$, then n is prime? (b) If $n\left|2^n-2\right.$, then If $m\left|2^m-2\right.$, where $m=2^n-1$; (c) For $n=161038$, then $n\left|2^n-2\right.$.

33. (a) n is a Carmichel number if $n\left|a^n-a\right.$ for any integer a. Show that 561 is a Carmichel number but 341 is not; (b) $m\geq 1$, $q_1=6m+1, q_2=12m+1$ and $q_3=18m+1$ are primes, then $q_1q_2q_3$ is a Carmichel number.

34. (a) If $n\left|2^n+2, n-1\right|2^n+1$, then $m\left|2^m+2, m-1\right|2^m+1$, where $m=2^n+2$. (b) Find infinitely many such numbers.

35. There exist infinitely many composite (not prime) numbers n such that for any integer a, $n\left|a^{n-1}-a\right.$.

36. Are $\log_2 10, \log_3 7, \log_{15} 21$ be rational? (No: by contradiction).

37. Let m and n be positive integers, show that $n!(m!)^n\left|(mn)!\right.$.

38. Let m and n be positive integers, show that $m!n!(m+n)!\left|(2a)!(2b)!\right.$.

39. Let n be composite, find the GCD of $\binom{n}{1},\binom{n}{2},...,\binom{n}{n-1}$.

40. (Beatty) Let α and β be real numbers. Define $\alpha_n = [\alpha n]$, $\beta_n = [\beta n]$, $n = 1, 2, 3, \ldots$. Show that the numbers are distinct and form the set of all positive integers iff α and β irrational with $\frac{1}{\alpha} + \frac{1}{\beta} = 1$.

41. Show that there exist infinitely many different prime numbers p and q such that $p \mid 2^{q-1} - 1$ and $q \mid 2^{p-1} - 1$.

42. Find all positive integers n such that $51^n - 2^n$ is a perfect square. (SAMC 2011, p.30)

43. For all positive integers n define $a_n = 6\underbrace{77\ldots7}_{n \text{ times}}$, where the digits 3 occurs n times.

 Show that the number a_{2012} has infinitely many multiples in the sequence $\{a_n\}$.
 (Rom 2009, p. 90)

44. Let $n \geq 10^{2012}$ be an integer. Find the first digit after the decimal point of $\sqrt{n^2 + 3n + 400}$. (Argentina 2010)

45. Find all positive integers n such that there exist n consecutive positive integers whose sum is a perfect square.

Diophantine Equations

1. Find all primes p and q such that $p^2 - 2q^2 = 1$.

2. How many pairs (m, n) of positive integers are there, for which $7m + 3n = 10^{2006}$ and m divides n?

3. Let $[x]$ stands for the largest integer smaller or equals to x. Solve $x^3 - [x] = 4$.

4. How many pairs of positive integers (x, y) satisfy the equation $x^2 + y^2 - 2004x - 2004y + 2xy - 2005 = 0$? (Answer: 2004)

5. Can you find integers x and y satisfying $15x^2 - 7y^2 = 9$?

6. Can you find non-zero integers x, y and z satisfying $x^2 + y^2 + z^2 = 2xyz$?

7. Determine all non-negative integral pairs (x, y) such that $(xy - 7)^2 = x^2 + y^2$.

8. Find all triples (a, b, c) of integers such that $a + b + c = 2018 \times 2019$ and the solutions to the equation $2019x^3 + ax^2 + bx + c = 0$ are all nonzero integers. (SAMC 2011)

9. Find all primes p and all positive integers a and m such that $a \leq 5p^2$ and $(p-1)! + a = p^m$.

Miscellaneous

1. John writes a number with 2187 digits on the blackboard, each digit being a 1 or 2. Judith creates a new number from John's number by reading the number from left to right and whenever she sees a 1 writing 112 and wherever she sees a 2 writing 111. (For

example, if John's number begins 2112, then Judith's number would begin 111112112111.) After Judith finishes writing her number, she notices that the left most 2187 digits in her number and in John's number are the same. How many times do five 1's occur consecutively in John's number?

2. A set of positive integers has the property that

Every member in the set, apart from 1, is divisible by at least one of 2, 3, or 5.

If the set contains $2n, 3n$ or $5n$ for some integer *n*, then it contains all three and *n* as well.

The set contains between 300 and 400 numbers. Exactly how many does it contain?

A. 324 B. 343 C. 351 D. 360 E. 364

3. Given a rectangular grid of *m* units wide and *n* units tall, where *m* and *n* are positive integers with $2n < m < 3n$. The region below the line linking (0, 0) and (*m*, *n*) is shaded. There is a unit square whose shaded area is greater than 0.999 (but not totally shaded). The smallest possible value of *mn* satisfies

A. $496 \le mn \le 500$ B. $501 \le mn \le 505$ C. $506 \le mn \le 510$ D. $511 \le mn \le 515$
E. $516 \le mn \le 520$

4. How many odd coefficients are there in the expansion of $(x^2 - x + 1)^{2009}$?

5. Let *n* be a positive integer and let *a* be an integer with $\gcd(a, n) = 1$.

Prove that

$$\frac{a^{\phi(n)} - 1}{n} \equiv \sum_{\substack{j=1 \\ \gcd(j,n)=1}}^{n} \frac{1}{aj}\left[\frac{aj}{n}\right] \pmod{n},$$

where $\phi(n)$ is the Euler's totient function and $[x]$ is the greatest integer not exceeding *x*.

6. Let *p* be an odd prime such that $p \equiv 1 \pmod 4$. Evaluate with reasons, $\sum_{k=1}^{\frac{p-1}{2}} \left\{\frac{k^2}{p}\right\}$, where $\{x\} = x - [x]$, $[x]$ being the greatest integer not exceeding *x*.

Quadratic Congruence

1. $(a, p) = 1$ and p an odd prime of the form $4k + 3$. Show that the solutions of $x^2 \equiv a \pmod p$ are $\pm a^{k+1}$.

2. quadratic residue of the odd prime *p*, show that $-a$ is a quadratic residue modulo *p* iff $p \equiv 1 \pmod 4$.

3. (a) *p* an odd prime, *q* the smallest positive quadratic nonresidue modulo *p*, then *q* is prime. (b) *p* an odd prime, *q* the smallest positive nonresidue modulo *p*, then $q < \sqrt{p} + 1$.

4. There exist infinitely many primes of the form $4k + 1$.

5. $83 \mid 2^{41}-1$?

6. $1999 \mid 2^{999}-1$?

7. q is odd and $p=4q+1$ is prime. (i) 2 is a quadratic nonresidue of p. (ii) $p \mid 4^{q}+1$.

8. Let p be an odd prime, find the number of quadratic nonresidue of p^{n}.

9. 3 is a nonresidues for all primes of the from $4^{n}+1$.

10. p a prime of the form $8k+3$, $p \,\Big|\, 2^{\frac{p-1}{2}}-1$?

11. $1!+2!+...+n!$ is never a square if $n>3$.

12. There exist infinitely many primes of the form (i) $8k+3$, (ii) $8k+5$ and (iii) $8k+7$.

Orders, Primitive Roots

1. 6 is a primitive root modulo 41, find the least positive residues of all elements of order 8 modulo 41.
2. If p is an odd prime, then the order of a modulo p is 2 iff $a \equiv -1 \pmod{p}$.
3. g a primitive root modulo p. If $p \equiv 3 \pmod{4}$, then $-g$ is of order $\dfrac{p-1}{2}$.

4. $a_1, a_2, ..., a_r$ a reduced residue system modulo m, then $a_1^n, a_2^n, ..., a_r^n$ a reduced residue system modulo m iff $(n, \phi(m))=1$.

5. Solve $x^5 \equiv 6 \pmod{101}$.

6. m is a positive integer having a primitive root, $(a,m)=1$, then $x^k \equiv a \pmod{m}$ has a solution iff $a^{\frac{\phi(m)}{(k,\phi(m))}} \equiv 1 \pmod{m}$. Moreover if $x^k \equiv a \pmod{m}$ has a solution, then it has $(k, \phi(m))$ of them.

7. $p=3k+2$, then a is a cubic residue.

8. $p=3k+1$, there are precisely one-third of the integers $1, 2, ..., p-1$ are cubic residues.

9. Find all primitive roots modulo 41.

10. Which integer has exactly one primitive root?

11. g a primitive root of m, the quadratic residues are even powers of g and nonresidues odd powers of g.

12. If g^k a primitive root of m, then g is a primitive root modulo m.

13. -3 is not a primitive root of any prime of the form 4^n+3.

14. Find all prime numbers p and q for which $a^{3pq} \equiv a \pmod{3pq}$ for all integers a.

15. Let m be a positive integer. (a) Prove that if $2^{m+1}+1$ divides $3^{2^m}+1$, then $2^{m+1}+1$ is a prime. (b) Is the converse true?

March 2017 Revised September. 2018

Divisibility

Some Divisibility Tests (*N* = 114043, in fact it is a prime number)

Divisors	Reduction	Reason
$7, 11, 13$	$114-043=71$	$7\times 11\times 13=1001$
37	$114+043=157$	$37\times 27=999$
$23, 29$	$114-2(043)=28$	$23\times 29\times 3=2001$
$31, 43$	$114+4(043)=286$	$31\times 43\times 3=3999$
19	$1140+4(43)=1312$	$19\times 21=399$
$17, 47$	$1140+8(43)=1484$	$17\times 47=799$
41	$1140+16(43)=1828$	$41\times 39=1599$
67	$1140-2(43)=1054$	$67\times 3=201$
89	$1140-8(43)=796$	$89\times 9=801$
53	$1140-9(43)=753$	$53\times 17=901$

1. Find the GCD of 952 and 4301.
1. Answer: 17.
2. The number $\overline{65xy13}$ is divisible by 99, find all positive value(s) of $x+y$.
2. Answer: 3.
3. The six-digit number $\overline{a2006b}$ is divisible by 72, find *a* and *b*.
3. Answer: 6, 4.
4. Find the unit digit of the number $1!+2!+3!+...+2009!$. (Note: $n!=n(n-1)(n-2)...1$.)
4. Answer: 3.
5. If the 9-digit number $\overline{123xyz789}$ is divisible by 999, find the value of $x+y+z$.
5. Answer: 15.
6. What is the remainder when $1^3+2^3+\cdots+2009^3$ is divided by 7?
6. Answer: 0
7. Find the remainder when $1^3+6^3+11^3+16^3+\cdots+2001^3$ is divided by 2002.
7. Answer: 1001.
8. $n+3$ divides $n^3+3n+2006$ and n is a 3-digit number. Determine the largest possible value of *n*.
8. Answer: 982.
9. Find the smallest positive integer *n* such that $999999n=111...11$.
9. We need $(10^6-1)n=\dfrac{10^k-1}{9}$, hence $n=\dfrac{10^k-1}{9(10^6-1)}$ with $k=6m$. Then $n=\dfrac{(1+10^6+...+10^{6(m-1)})}{9}$, the numerator becomes a multiple of 9 if $m=9$.
10. Find the last two digits of $14^{14^{14}}$.

10. Note the number $\equiv 0 \pmod 4$. Now $14^{14} \equiv 4^{14} \equiv (4^2)^7 \equiv 6 \pmod{10}$. Hence $14^{14^{14}} = 14^{10t+6} = 14^{10t}14^6 \equiv 14^6 \equiv -14 \equiv 11 \pmod{25}$. The last two digits of the number is 36.
11Does it exist a multiple of 2009 of the form $11\ldots1$?
11. Yes.
12. Let n be an odd integer, show that the last two digits of $2^{2n}(2^{2n+1}-1)$ is 28.
12. $2^{4m+2}(2^{4m+3}-1)-28 = 4(2^{8m+3}-2^{4m}-7) = 4(8\times(2^{4m})^2-2^{4m}-7)$
$= 4(2^{4m}-1)(8\times 2^{4m}+7) = 4(16^m-1)(8(16^m-1)+15)$.
Check that 16^m-1 is divisible by 5.
13. Let $a_1, a_2, \cdots a_n$ be a permutation (rearrangement without omission nor repetition) of 1, 2, …, n. If n is odd, show that $(a_1-1)(a_2-2)\cdots(a_n-n)$ is even. Is the claim true if n is even?
13. The number is odd only when all $a_i - i$ odd!
14. If p is a prime number, show that $C_i^p = \dfrac{p!}{i!(p-i)!}, 1 \le i \le p-1,$ is always divisible by p. (You may assume C_i^p is always a positive integer.) Is the claim true if p is not a prime number?
14. Note $C_i^p i!(p-i)! = p!$, the right-hand side is divisible by p, while $i!$ and $(p-i)!$ are not.
15. Show that (a) $10n+b$ is divisible by 7 if and only if $n+5b$ is, (b) $10n+b$ is divisible by 13 if and only if $n+4b$ is, (c) $10n+b$ is divisible by 17 if and only if $n+12b$ is.
15. (a) 7 divides $10n + b$ if and only if it divides $50n + 5b$, if and only is it divides $n + 5b$. Similarly one proves (b) and (c).
16. Let n be an odd positive integer. Then the largest positive integer k, such that $n^{12}-n^8-n^4+1$ is divisible by 2^k for every n is A. 6 B. 7 C. 8 D. 9 E. 10?
16. $n^{12}-n^8-n^4+1 = (n^8-1)(n^4-1) = (n^4+1)(n^2+1)^2(n+1)^2(n-1)^2$. Take $n \equiv 1, 3 \pmod 4$, we find altogether nine 2s.
17. How many positive integers n are there such that $8n+50$ is a multiple of $2n+1$?
17. Answer: 1.
18. Given two relatively prime natural numbers a and b, show that the greatest common factor of $a+b$ and a^2+b^2 is 1 or 2.
18. $d = GCD(a+b, a^2+b^2)$, then $d \mid 2ab$, hence $2a^2 = 2a(a+b)-2ab$ is divisible by d, and so is $2b^2 = 2b(a+b)-2ab$.
19. Find the least integer n with the following property: given n distinct integers $a_1, a_2, ..., a_n$, there are two whose sum or difference is divisible by 9.
19. Answer: 6
20. Let p be a prime number. Show that there exists a prime number q such that for every integer n, the number $n^p - p$ is not divisible by q.

20. (IMO2003) We have $\frac{p^p-1}{p-1}=1+p+p^2+...+p^{p-1}\equiv p+1 \pmod{p^2}$, hence we can get at least one prime divisors of $\frac{p^p-1}{p-1}$ which is not congruent to 1 mod p^2. Denote such a prime divisor q. This q is what we want. Assume there is an integer n such that $n^p \equiv p \pmod q$, then $n^{p^2}\equiv p^p \equiv 1 \pmod q$ by the definition of q. On the other hand, by the Fermat's little theorem, we have $n^{q-1}\equiv 1 \pmod q$. Since p^2 does not divide $q-1$, we have $\gcd(p^2, q-1)$ divided p, which leads to $n^p \equiv 1 \pmod q$. Hence we have $p \equiv 1 \pmod q$. However this implies $1+p+...+p^{p-1}\equiv p \pmod q$. From the definition of q, this leads to $p\equiv 0 \pmod q$, a contradiction.

21. Let $n \ge 0$, define $F_n = 2^{2^n}+1$. (Fermat numbers) (a) If $m \ne n$, $d \mid F_n$, then d does not divide F_m. Hence show there exist infinitely many primes. (b) $F_{n+1}=F_n...F_0+2$. (c) $A_1 = 2$, $A_{n+1}=A_n^2-A_n$, $n\ge 1$. Let $n \ne m$. If $1<d\mid A_n$, then d does not divide A_m. Hence show there exist infinitely many primes. (d) $A_{n+1}=A_n...A_1+1$.

21. (a) If $m<n$, then $F_n \mid 2^{2^m}-1$. (b) Induction. (c) If $m<n$, then $A_n \mid A_{m-1}$. (d) Induction.

22. If there are finitely many primes $p_1, p_2, ..., p_s$, then for any integer N, $\sum_{n=1}^{N}\frac{1}{n}<(1-\frac{1}{p_1})^{-1}...(1-\frac{1}{p_s})^{-1}$.

22. Indeed $\lim_{N\to\infty}\sum_{n=1}^{N}\frac{1}{n}=\infty$.

23. Let $N\ge 2$. (a) $\sum_{p<N}\frac{1}{p-1}>\ln\ln(N+1)$; (b) $\sum_{p<N}\frac{1}{p}>\ln\ln(N+1)-1$.

23. (a) Use $\frac{1}{n}>\ln(1+\frac{1}{n})$ and $\ln(1+\frac{1}{p-1})<\frac{1}{p-1}$; (b) $\sum_{p\le N}\left(\frac{1}{p-1}-\frac{1}{p}\right)<1$.

24. (a) Any positive integer a is of the form $k^2 l$, where l is a product of distinct primes or 1; (b) $N\ge 2$, then $N\le\sqrt{N}\cdot 2^{\pi(N)}$, where $\pi(x)$ is the number of primes less than or equal to x. (c) $\pi(N)\ge\frac{\log_2 N}{2}$; (d) The n^{th} prime $p_n\le 2^{2n}$.

24. (a) Unique factorization, Clear; (b) $1\le k\le\sqrt{N}$ and different possible values of $l<1+\binom{\pi(N)}{1}+...+\binom{\pi(N)}{\pi(N)}=2^{\pi(N)}$; (c), (d) Use (b).

25. Suppose 2^n-1 divides m^2+81 for some integer m, then n is a power of 2. Otherwise n has an odd divisor $l\ge 3$ and such that 2^l-1 divides 2^n-1 and hence m^2+81. But for $l\ge 3$, $2^l-1\equiv -1\equiv 3 \pmod 4$, hence 2^l-1 has a prime divisor $p\equiv 3 \pmod 4$. Since l is odd, $p\ne 3$. Now $p \mid m^2+81=m^2+9^2$, hence from the Fermat's little theorem,

$1 \equiv m^{p-1} \equiv (m^2)^{\frac{p-1}{2}} \equiv (-81)^{\frac{p-1}{2}} \equiv 9^{p-1}(-1)^{\frac{p-1}{2}} \equiv -1 (\bmod p),$
a contradiction.

Now let $n = 2^k$. The assertion is true if $k = 0$. Assume now $k \geq 1$. We have $2^n - 1 = 2^{2^k} - 1 = 3(2^2+1)(2^{2^2}+1)...(2^{2^{k-1}}+1)$. Note that if $r \neq s$, then $2^{2^r}+1$ and $2^{2^s}+1$ are relatively prime. Indeed if $d \geq 2$ is their greatest common divisor, and $r > s$, then $-1 \equiv 2^{2^r} \equiv (2^{2^s})^{2^{r-s}} \equiv (-1)^{2^{r-s}} \equiv 1 (\bmod d)$, a contradiction. Now by the Chinese Remainder Theorem, there exists a natural number c such that
$c \equiv 2^{2^j} (\bmod 2^{2^{j+1}}+1)$, with $j = 0, 2, ..., k-2$. Thus $c^2 + 1 \equiv 0 (\bmod 2^{2^{j+1}}+1)$, with $j = 0, 2, ..., k-2$. This implies $2^n - 1$ divides $(9c)^2 + 81$.

26. Let k be a positive integer such that $p = 8k+5$ is a prime number. The integer $r_1, r_2, ..., r_{2k+1}$ are chosen so that the numbers $0, r_1^4, r_2^4, ..., r_{2k+1}^4$ give pairwise different remainders modulo p. Prove that the product $\prod_{1 \leq i < j \leq 2k+1} (r_i^4 + r_j^4)$ is congruent to $(-1)^{k)k+1)/2}$ modulo p.

26. (3rd IOM) Let g be a primitive root modulo p such that $1, g, g^2, ..., g^{p-2}$ are all different nonzero remainders modulo p. The $2k+1$ nonzero fourth powers modulo p are $1, g^4, g^8, ..., g^{8k}$, congruent to $r_1^4, r_2^4, ..., r_{2k+1}^4$ in some order. Define the map $f(j): \{0,1,...,2k\} \to \{0,1,...,2k\}$ as a remainder of $2j$ modulo $2k+1$. Note $8j$ and $4f(j)$ are congruent modulo $4(2k+1) = p-1$, thus $g^{8j} \equiv g^{4f(j)}$ modulo p for all $j = 0, 1, 2, ..., 2k$. We have

$$\prod_{1 \leq i < j \leq 2k+1} (r_i^4 + r_j^4) = \prod_{1 \leq i < j \leq 2k+1} \frac{r_j^8 - r_i^8}{r_j^4 - r_i^4}$$
$$\equiv \prod_{1 \leq i < j \leq 2k+1} \frac{g^{8j} - g^{8i}}{g^{4j} - g^{4i}} \equiv \prod_{1 \leq i < j \leq 2k+1} \frac{g^{4f(j)} - g^{4f(i)}}{g^{4j} - g^{4i}} \quad (\bmod p).$$

We may write $g^{4f(j)} - g^{4f(i)} = \pm(g^{4\max(f(j),f(i))} - g^{4\min(f(j),f(i))})$, where the sign is positive if $f(j) > f(i)$ and negative if $f(j) < f(i)$. Further, when the ordered pair (i, j) runs over all $k(2k+1)$ ordered pairs satisfying $0 \leq i < j \leq 2k$, the ordered pair $(\min(f(j), f(i)), \max(f(j), f(i))$ run over the same set. Therefore the differences cancel out and the above ratios $\prod_{1 \leq i < j \leq 2k+1} \frac{g^{4f(j)} - g^{4f(i)}}{g^{4j} - g^{4i}}$ equals $(-1)^N$, where N is the number of pairs $i < j$ for which $f(i) > f(j)$. This in turn happens when $i = 1, 2, ..., k$; $j = k+1, ..., k+i$, totally $N = 1 + 2 + ... + k = k(k+1)/2$. Thus the result.

27. Let $1=d_0<d_1<...<d_m=4k$ be all positive divisors of $4k$, where k is a positive integer. Prove that there exists $i\in\{1,2,...,m\}$ such that $d_i-d_{i-1}=2$.

27. (3rd IOM) Assume not. This means if d and $d+2$ both divide $4k$, so is $d+1$. Note if a divides $4k$ and a not divisible by 4, then $2a$ divides $4k$. Start from the pair $(1,2)$ and find more pairs $(a,a+1)$ such that both divides $4k$ and both not divisible by 4.
Now from the initial observation, if $(a,a+1)$ is a pair both divide $4k$ but none divisible by 4, then so is the pair $(2a,2a+2)$, but now we have $2a+1$ also divides $4k$. One of $2a$ or $2a+2$ is not divisible by 4, hence in one of the pairs $(2a,2a+1),(2a+1,2a+2)$ both numbers divides $4k$, nut not divisible by 4.
Now applying this procedure from $(1,2)$, get $(2,3),(5,6),(10,11)$, etc. At each step the sum of numbers in pair increases, and we obtain an infinite set of divisors of $4k$, a contradiction.

Numbers

1. Show that $1+\frac{1}{2}+...+\frac{1}{n}, n\geq 2,$ is never an integer.
1. (a) Let $2^{q+1}\leq n<2^{q+2}$. Multiply the expression by $2^q P$, where P is the product of all odd numbers between 1 and n. (b) Similar proof.
2. Determine all pairs (m,n) of positive integers for which 2^m+3^n is a square.
2. By considering the number mod 3, get m even, by considering the number mod 4, get n even. Put $m=2r, n=2s$, have $2^{2r}=(a-3^s)(a+3^s)$, hence $2^i=a-3^s$ and $2^{2r-i}=a+3^s$. From these two expressions, get $2\times 3^s=2^i(2^{2r-2i}-1)$, hence $i=1$. Thus $3^s=2^{2r-2}-1$. $s>1$ is impossible by considering the expression mod 8, so $s=1, n=2$.
3. Prove that n^4+4^n is not a prime number for any integer $n>1$.
3. n is odd, hence $n^4+4^n=n^4+2^{2n}=(n^2+2^n)^2-2^{n+1}n^2$ and it can be further factorized.
4.Find a 4-digit square number such that when each digit is increased by 3, the resulting 4-digit number is still a square.
4. Answer: 1156.
5. Is $1\underbrace{00...0}_{2009\ 0's}1$ a prime?
5. No. Use sum of cubes.
6. If n is a positive integer and $n^2=11112222^2+3333^2+3334^2$, find n.
6. Answer: 11112223.
7. Show that for any natural number n, $\frac{21n+4}{14n+3}$ is irreducible (i.e. the fraction cannot be further reduced).
7. Show $GCD(21n+4,14n+3)=1$.
8. A positive integer n is *good* if the digits of n can be divided into two groups such that the sum of one group of the digits in one group is equal to the sum of the other, (for example, 1432 is good). Find the smallest possible n such that both n and $n+1$ are good.

8. Answer: 549.
9. Show that the product of four consecutive numbers is not a cube.
9. Consider $n(n+1)(n+2)(n+3)$. If n even, then $n+1$ is relatively prime to $n(n+2)(n+3)$. Hence both $n+1$ and $n(n+2)(n+3)$ are cube. But check $(n+1)^3 < n(n+2)(n+4) < (n+2)^3$. For n odd, check $n+2$ relatively prime to $n(n+1)(n+3)$.
10. Find the least number whose last digit is 7 and which becomes 5 times larger when this last digit is carried to the beginning of the number.
10. The number is $\ldots abc7$, then $5(...abc7) = 7...abc$, hence $c = 5$. Put c back to get $b = 8$. Continuing in this manner to get 7 for the first time, and the number is 142857.
11. There are five positive integers. Taken four at a time, they sum to four distinct values, namely 14, 15, 19 and 20. Find the sum of the five numbers.
11. Answer: 22.
12. A number n equals to the sum of squares of its four smallest divisors, find the largest prime factor of n.
12. Answer: 13. By considering the problem mod 4, n must be equal and not divisible by 4, hence 1 and 2 are the smallest factors of n. Let p be the next odd prime factor, then the next smallest factor is $2p$, get $p = 5$. Hence $n = 130$.
13. For any m, there exist infinitely many n such that $m^4 + n$ is composite.
13. Take $n = 4k^4$, completing squares.
14. Can you find a natural number n such that $n!$ ends in exactly 2009 zeros?
14. We need to find n such that 5^{2009} is the maximum power of 5 that divides $n!$. Solve $2009 = \left[\frac{n}{5}\right] + \left[\frac{n}{25}\right] + \left[\frac{n}{125}\right] + ... \le \frac{n}{5} + \frac{n}{25} + \frac{n}{125} + ... \le \frac{n}{4}$, get $n \ge 8036$. Try 8036 to get 5^{2006}, need three more 5s, hence to 8045, it works.
15. Find all 4-digit numbers $\overline{aabb}$ which is a perfect square.
15. Answer: 7744. $\overline{aabb} = 11(100a + b)$, need $a + b \equiv 0 \pmod{11}$ and the factor $\frac{100a+b}{11}$ a square.
16. It is known that 999973 has *exactly* three distinct prime factors. Find the sum of these three prime factors.
16. We have

$$\begin{aligned}
999973 &= 1000000 - 27 \\
&= 100^3 - 3^3 \\
&= (100-3)(100^2 + 100\times 3 + 3^2) \\
&= 97\times 10309 \\
&= 97\times 13\times 793 \\
&= 97\times 13^2\times 61
\end{aligned}$$

and so the answer is $97 + 13 + 61 = 171$.

17. Find the 4-digit number $\overline{abcd}$ such that $4\overline{abcd} = \overline{dcba}$.

17. $a<3$. Also, the number must be even, so $a=2$. From $4\overline{2bcd}=\overline{dcb2}$, we must have $d=8$. Expand to get $13b+1=2c$. The right side is even and b must be odd. Also $2c\le 18$, get $b=1, c=7$.

18. If $2^8+2^{11}+2^n$ is a perfect square, find n.

18. $2^8+2^{11}=2^8 3^2=48^2$. Thus if $2^8+2^{11}+2^n=k^2$, then $2^n=(k+48)(k-48)$, get $k-48=2^a$ and $k+48=2^b$. Get $k=80$, $n=12$.

19. Find the smallest multiple of 63 which contains only the digits 8 or 3.

19. Answer: 8883.

20. Find the smallest multiple of 2834 which contains only the digits 0, 1 or 9.

20. Answer: 99190.

21. What is the unit digit of $(2+1)(2^2+1)(2^4+1)(2^8+1)(2^{16}+1)(2^{32}+1)(2^{64}+1)$? (Use factorization of a^2-b^2.)

21. Answer: 5.

22. If x, y, z and n are natural numbers with $x^n+y^n=z^n$, then show that x, y and z are all greater than n.

22. Let $z>y\ge x$, thus $z\ge y+1$, get

$x^n=z^n-y^n=(z-y)(z^{n-1}+z^{n-2}y+\ldots+y^{n-1})>((y+1)-y)nx^{n-1}$. Thus $x>n$.

23. If $\frac{1}{a}+\frac{1}{b}=\frac{1}{c}$ where a, b, c are positive integers with no common factor, show that $a+b$ is a square.

23. We must have $a>c$ and $b>c$. Put $a=c+m$ and $b=c+n$ back to the equation and simplify, get $c^2=mn$. Now note that m, n are relatively prime, otherwise a number d (larger than 1) and it divides both m and n, so is c, and so is a and b. Thus we have $m=k^2, n=l^2$. Put these back into the equation to get $a+b=(k+l)^2$.

24. Start from Machine A, (m, n) becomes (n,m); while Machine B and C will produce $(m+3n,n)$ and $(m-2n,n)$ respectively. Start from (0, 1) and put into anyone of the machines, after finitely many rounds, which of the following pairs cannot be produced?

A. (2009,1016) B. (2009,1004) C. (2009,1002) D. (2009,1008)

E. (2009,1032)

24. (Pascal 2009) D. GCD is an invariant

25. Let p be a prime of the form $3k+2$. Show that any number modulo p is a cube.

25. $(3,p-1)=1$, hence there exist s and t such that $3s+t(p-1)=1$. Therefore $a^1=a^{3s+t(p-1)}=a^{3s}(a^{p-1})^t\equiv(a^s)^3 \pmod p$.

26. Show that for any integer $n>1$, $\frac{2^n-1}{n}$ is not an integer. (Use Fermat's little theorem and consider the smallest prime factor of n.

26. Let p be the smallest prime factor of n, it is odd. Hence $2^n\equiv 1 \pmod p$. Likewise $2^{p-1}\equiv 1 \pmod p$. Thus $2^{GCD(p-1,n)}\equiv 1 \pmod p$. Yet $GCD(p-1,n)=1$, a contradiction.

27. There exist infinitely many n such that n divides 2^n+2. Find one between 100 and 2000.

27. Suppose $n \mid 2^n+2$ and $n-1 \mid 2^n+1$. Take $m=2^n+2$, consider 2^m+2 and 2^m+1.
Answer: 946.

28. Find one pair of positive integers a, b such that (1) $ab(a+b)$ is not divisible by 7; (2) $(a+b)^7-a^7-b^7$ is divisible by 7^7.

28. (IMO1984) Expand $(a+b)^7-a^7-b^7=7ab(a+b)(a^2+ab+b^2)^2$. Hence 7^3 divides a^2+ab+b^2. This implies $a^3 \equiv b^3 \pmod{7^3}$. Take $b=1$, then $a^3 \equiv 1 \pmod{7^3}$. But $2^{\phi(7^3)}=2^{294} \equiv (2^{98})^3 \equiv 1 \pmod{7^3}$, we may take $a=2^{98}$. (In fact $a=18$ will also do.)

29. (a) Show that 3^{n+1} just divides $2^{3^n}+1$ by induction.
(b) Find an integer n, where n is not purely a power of 3, and n divides 2^n+1.
(c) Is it possible to find n with exactly 2009 prime divisors, and such that n divides 2^n+1.

30. The real number $0.a_1a_2a_3...$ is defined as follows, a_n is the first non-zero digit (counted from the right) of $n!$. For instance, $1!=1$, hence $a_1=1$. Likewise $a_2=2$, $a_3=6$, $a_4=4$, $a_5=2$ since $5!=120$. Is the number rational?

30. If the number is rational then it is eventually periodic, say starting from N, $a_{N+i+T}=a_{N+i}$, here T is the period. Suppose $T=2^r5^sq$, with $(q,10)=1$. By the Euler's theorem we can find an integer t such that $10^t \equiv 1 \pmod{q}$. We can also find an integer n such that 2^r5^s divides 10^n. Thus T divides $10^n(10^t-1)$. Let $m=n+t$, then

$$10^m \equiv 10^n \pmod{T}. \quad (1)$$

By multiplying the expression with powers of 10 we may assume $m>n>T$. Observe that $(10^m)!=10^m(10^m-1)!$, hence the first nonzero digits of $(10^m)!$ and $10^m(10^m-1)!$ are the same. This implies

$$a_{10^m}=a_{10^m-1} \quad (2)$$

From this and by the periodicity we have $a_{10^m+(10^m-10^n)}=a_{10^m+(10^m-10^n)-1}$, or

$$a_{2\times10^m-10^n}=a_{2\times10^m-10^n-1} \qquad (3)$$

Observe now the first non-zero digit of $2\times10^m-10^n$ is 9, and $(2\times10^m-10^n)!=(2\times10^m-10^n)(2\times10^m-10^n-1)!$. If $a_{2\times10^m-10^n-1}$ equals s, then by (3) and the above observation we must have $9s \equiv s \pmod{10}$, or $s=5$. But this is impossible, for a_i never equals 5. (In fact if $n!=2^{t_2}3^{t_3}5^{t_5}...$, then $t_2=\sum_{j=1}^{\infty}\left[\frac{n}{2^j}\right] \geq \sum_{j=1}^{\infty}\left[\frac{n}{5^j}\right]=t_5$, which means the 5s in $n!$ will always be cancelled out by the 2s.) This concludes the proof that the number is irrational.

31. Given positive integers a and b and such that $ab+1$ divides a^2+b^2, then $\frac{a^2+b^2}{ab+1}$ is a perfect square.

31. (IMO1998) Let $\frac{a^2+b^2}{ab+1}=k$ and such that k is not a perfect square. After re-arranging we have $a^2-kab+b^2=k$, with $a>0$ and $b>0$. Assume now (a_0,b_0) is a solution of the Diophantine equation and such that a_0+b_0 is as *small* as possible. By symmetry we may assume $a_0 \ge b_0 > 0$. Fixing b_0 and k, we may assume a_0 is a solution of the quadratic equation $x^2-kb_0x+b_0^2-k=0$. Now let another root of the equation be a'. Using sum and product of roots, we have $a_0+a'=kb_0$ and $a_0a'=b_0^2-k$. The first equation implies a' is an integer. The second equation implies $a'\neq 0$, otherwise k is a perfect square, contradicting our hypothesis. Now a' also cannot be negative, otherwise $a'^2-ka'b_0+b_0^2 \ge a'^2+k+b_0^2 > k$. Hence $a'>0$. Finally $a'=\frac{b_0^2-k}{a_0} \le \frac{b_0^2-1}{a_0} \le \frac{a_0^2-1}{a_0} < a_0$. This implies (a',b_0) is an integer solution of $a^2-kab+b^2=k$, and $a'+b_0 < a_0+b_0$, contradicting the minimality of a_0+b_0. Hence k must be a perfect square. □

32. (a) If $n|2^n-2$, then n is prime? (b) If $n|2^n-2$, then If $m|2^m-2$, where $m=2^n-1$; (c) For $n=161038$, then $n|2^n-2$.

32. (a) False, let $n=341$ (pseudo-prime); (b) let $2^n-2=nk$, then $2^{2^n-1}-2=2(2^{nk}-1)=2A(2^n-1)$. (c) $161038=2\cdot 73\cdot 1103$, $161038-1=3^2\cdot 29\cdot 617$; $73|2^9-1, 1103|2^{29}-1$.

33. (a) n is a Carmichel number if $n|a^n-a$ for any integer a. Show that 561 is a Carmichel number but 341 is not; (b) $m\ge 1$, $q_1=6m+1, q_2=12m+1$ and $q_3=18m+1$ are primes, then $q_1q_2q_3$ is a Carmichel number.

33. 31 does not divide $11^{341}-1$; (b) Let $n=q_1q_2...q_k$ with $q_i s$ distinct primes such that $q_i-1|n-1$, all i, then n is Carmichel; $541=3\cdot 11\cdot 17$.

34. (a) If $n|2^n+2, n-1|2^n+1$, then $m|2^m+2, m-1|2^m+1$, where $m=2^n+2$. (b) Find infinitely many such numbers.

34. (Sierpenski) (a), (b) Start from $n=2$.

35. There exist infinitely many composite (not prime) numbers n such that for any integer a, $n|a^{n-1}-a$.

35. $n=2p$, where p is prime.

36. Are $\log_2 10, \log_3 7, \log_{15} 21$ be rational? (No: by contradiction).

37. Let m and n be positive integers, show that $n!(m!)^n|(mn)!$.

38. Let m and n be positive integers, show that $m!n!(m+n)! \mid (2a)!(2b)!$.

39. Let n be composite, find the GCD of $\binom{n}{1}, \binom{n}{2}, \ldots, \binom{n}{n-1}$.

39. $n = p^k$, then GCD = p, note $p \,\Big\|\, \binom{p^k}{p^{k-1}}$. n not of the form, GCD = 1.

40. (Beatty) Let α and β be real numbers. Define $\alpha_n = [\alpha n], \beta_n = [\beta n], n = 1, 2, 3, \ldots$. Show that the numbers are distinct and form the set of all positive integers iff α and β irrational with $\frac{1}{\alpha} + \frac{1}{\beta} = 1$.

41. (Mongolian 2008) Let $F_n = 2^{2^n} + 1$. Now all prime divisors of F_n are equal to 1 modulo 2^{n+1}. Indeed $2^{2^n} \equiv -1 \pmod p$, thus order of 2 modulo p is 2^{n+1} and it divides $p-1$. Now we show for each F_n we can find p and q satisfying the required conditions and one of p and q is a factor of F_n. As these numbers are co-prime, we got an infinite list. Indeed suppose $F_n = 2^{2^n} + 1 = p^k, k \geq 2$, then $p - 1 = 2^u$ and $\frac{p^k - 1}{p-1} = 2^v$ for some u and v. The congruence $2^v = p^{k-1} + p^{k-2} + \ldots + 1 \equiv k \pmod{2^u}$ implies that k is divisible by 2^u, so $k \geq 2^u$. By order argument, $u \geq n+1$. Hence $p^k \geq p^{2^u} \geq p^{2^{n+1}} > 2^{2^{n+1}} < F_n$, a contradiction. Thus F_n is either prime or has at least two distinct prime factors. In the first case, $p = F_n$ and some prime factor q of F_{n+1} will satisfy our conditions. Indeed $q - 1 = 2^{n+2} m$. Hence $2^{q-1} = (2^{2^n})^{4m} \equiv (-1)^{4m} \equiv 1 \pmod p$ and
$2^{p-1} = 2^{2^{2^n}} = (2^{2^{n+1}})^{2^{2^n - n - 1}} \equiv 1 \pmod q$, because $2^n > n+1$ for $n > 1$. In the second case, any two prime divisors p and q will do. Indeed $p - 1 = 2^{n+1} m$ and $q - 1 = 2^{n+1} k$. Check the congruence relations are valid.

42. Find all positive integers n such that $51^n - 2^n$ is a perfect square. (SAMC 2011, p.30)

When $n = 1$, $51 - 2 = 49 = 7^2$. We show that there are no more squares. Indeed if $n = 2k+1$ is odd, with $k \geq 1$, then

$$51^{2k+1} - 2^{2k+1} = (52-1)^{2k+1} - 2 \times 2^{2k}$$
$$= (52-1)^{2k+1} - 2 \times 4^k$$
$$= 4m + (-1)^{2k+1} \equiv -1 \pmod 4,$$

And it cannot be a perfect square.

If $n = 2k$ is even, then $(51^k - 1)^2 < 51^{2k} - 2^{2k} < (51^k)^2$, hence $51^{2k} - 2^{2k}$ cannot be a perfect square.

43. For all positive integers n define $a_n = 6\underbrace{77...7}_{n\,\text{times}}$, where the digits 3 occurs n times. Show that the number a_{2012} has infinitely many multiples in the sequence $\{a_n\}$.

(Rom 2009, p. 90)

43. The number a_{2012} is not divisible by 2 or 5, hence it has a multiple M with all digits equal to 1. Let $m > 0$ be the number of digits of M, i.e. $M = \underbrace{11...1}_{m\,\text{times}}$. Then $a_{2012+m} = a_{2012} \times 10^m + 7M$ and it is divisible by a_{2012}. Proceed accordingly.

44. Let $n \geq 10^{2012}$ be an integer. Find the first digit after the decimal point of $\sqrt{n^2+3n+400}$.

(Argentina 2010).

44. The answer is 5. It is easy to guess by noticing $\lim_{n\to\infty}(\sqrt{n^2+3n+40}-(n+\frac{3}{2}))=0.$ Actually the answer is 5 for $n \geq 2000.$ If $n \geq 2000,$ we have $(n+1)^2 < n^2+3n+400 < (n+2)^2$, and $\left[n^2+3n+400\right] = n+1.$ Hence $\sqrt{n^2+3n+400}$ is irrational. Let k be the first digit of $\sqrt{n^2+3n+400}$ after the decimal point, with $0 \leq k \leq 9.$ Then

$n+1+\frac{k}{10} < \sqrt{n^2+3n+400} < n+1+\frac{k+1}{10},$ or

$10n+10+k < 10\sqrt{n^2+3n+400} < 10n+10+(k+1).$

The inequality is strict as $\sqrt{n^2+3n+400}$ is irrational. Squaring and simplifying to get $20(n+1)k+k^2 < 100n+100+39800 < 20(n+1)(k+1)+(k+1)^2.$ The left inequality implies $(n+1)(k-5) < 199.$ As $n \geq 2000,$ need $k \leq 5.$ The right inequality implies $39800 < 20(n+1)(k-4)+(k+1)^2 < 20(n+1)(k-4)+10^2.$ As $n \geq 2000,$ we need $k \geq 5.$ Together we get the result.

45. Find all positive integers n such that there exist n consecutive positive integers whose sum is a perfect square.

45. (1st IOM). $n = 2^s m,$ where m is any odd integer and s is 0 or odd. Let $S(n,t) = (t+1)+(t+2)+...+(t+n) = (2t+n+1)n/2.$ For n odd, let $t = (n-1)/2$ to get $S(n,t) = n^2.$ For n even, $n = 2^s m,$ where s is positive and m odd, thus $2t+n+1$ is odd. Hence 2^{s-1} divides $S(n,t),$ while 2^s does not. This means for even s the answer is negative. For odd s one puts $t = (mx^2-n-1)/2$ for some odd $x > n$ and obtain $S(n,t) = 2^{s-1}m^2x^2.$

Diophantine Equations

1. Find all primes p and q such that $p^2 - 2q^2 = 1$.
1. Answer: (3, 2). p must be odd. Suppose both greater than 3, then both are $\pm 1 \pmod 6$, the equation will imply a contradiction.
2. How many pairs (m, n) of positive integers are there, for which $7m + 3n = 10^{2006}$ and m divides n?
2. Write $n = mk$, get $m(7+3k) = 10^{2006}$. The problem is reduced to finding $7 + 3k = 2^r 5^s$. First consider $r, s \ge 1$, taking mod 3, get $2^{r+s} \equiv 1 \pmod 3$, or $r + s$ is even. For each r between 1 and 2006, get 1003 choices of s, hence 2012018. If $r = 0$, get $s = 2, 4, ...,$ (1003 choices). If $s = 0$, get $r = 4, 6, 8, ...$ (1002 choices). Hence altogether the answer is 2014023.

3. Let $[x]$ stands for the largest integer smaller or equals to x. Solve $x^3 - [x] = 4$.
3. Answer: $\sqrt[3]{5}$.
4. How many pairs of positive integers (x, y) satisfy the equation $x^2 + y^2 - 2004x - 2004y + 2xy - 2005 = 0$?
4. Answer: 2004
5. Can you find integers x and y satisfying $15x^2 - 7y^2 = 9$?
6. Can you find non-zero integers x, y and z satisfying $x^2 + y^2 + z^2 = 2xyz$?
7. Determine all non-negative integral pairs (x, y) such that $(xy-7)^2 = x^2 + y^2$.
7. Obvious solutions (0, 7), (7, 0). If both x and y non-zeros, reduce the equation to $(xy-6)^2 - (x+y)^2 = 13$, get $(x, y) = (3,4), (4,3)$.
8. Find all triples (a, b, c) of integers such that $a + b + c = 2018 \times 2019$ and the solutions to the equation $2019x^3 + ax^2 + bx + c = 0$ are all nonzero integers.

SAMC 2011

8. Indeed consider an equation of the form $px^3 + ax^2 + bx + c = 0$, where $a + b + c = p(p-1)$. Let x_1, x_2, x_3 be its roots. Then $x_1 + x_2 + x_3 = -\frac{a}{p}$,

$x_1x_2 + x_2x_3 + x_3x_1 = \frac{b}{p}$ and $x_1x_2x_3 = -\frac{c}{p}$. Hence

$$(x_1 - 1)(x_2 - 1)(x_3 - 1) = -\frac{c}{p} - \frac{b}{p} - \frac{a}{p} - 1 = -\frac{a+b+c}{p} - 1 = -p.$$

Because $x_1, x_2, x_3 \ne 0$, so we must have $x_1 = 2$, $x_2 = 2$, $x_3 = 1 - p$, up to permutations. It follows that

$$5 - p = -\frac{a}{p}, 4(2-p) = \frac{b}{p}, 4(1-p) = -\frac{c}{p}.$$

Hence all triples are $(p(p-5), 4p(2-p), 4p(1-p))$.

In our case $p=2019$ and we obtain
$(a,b,c)=(2014\times 2019,-4\times 2017\times 2019,-4\times 2018\times 2019).$

9. Find all primes *p* and all positive integers *a* and *m* such that $a\le 5p^2$ and $(p-1)!+a=p^m$.

9.

$$(p,a,m)=(2,1,1),(2,3,2),(2,7,3),(3,1,1),(3,7,2),(3,25,3),$$
$$(5,1,2),(2,15,4),(5,101,3).$$

Direct checks give the above solutions. For $p\ge 7$, by Wilson $a\equiv 1 \pmod p$. Moreover $(p-1)\big|(a-1)$. Hence $a=kp(p-1)+1$ for some integer $k\ge 0$. If $k\ge 6, a\ge 6p^2-6p+1>5p^2$, a contradiction. Hence $0\le k\le 5$. Also after dividing by $p-1$, we obtain $(p-2)!+kp=p^{m-1}+p^{m-2}+...+1$. Since $p\ge 7$, then 2 and $(p-1)/2$ are different and appear in $(p-2)!$, i.e., $(p-1)\big|(p-2)!$. This gives $k\equiv m(\text{mod}(p-1))$. If $m\ge p$, then $(p-2)!+kp=p^{m-1}+p^{m-2}+...+1\ge p^{m-1}>(p-1)!$, which gives a contradiction. Thus $m\le p-1$. If $k=0$, then $m=p-1$ and
$(p-2)!=p^m-1\ge p^{p-1}-1>(p-1)^{p-1}>(p-1)!$, a contradiction.
If $k=1$ or $k=2$, then $m=1$ or 2, impossible.
If $k=3$, then $m=3$, thus $(p-2)!=(p-1)^2$, impossible.
For $k=4$ we get $m=4$ and $(p-2)!=(p-1)(p^2+p+1)$, and so $p-2$ divides $p^2+2p-1=p^2-4+2(p-2)+7$, or $p-2=7$, impossible.
For $k=5, m=5$ and $(p-2)!=(p-1)(p^3+2p^2+3p-1)$ and so $p-2$ divides $p^3+2p^2+3p-1=p^3-8+2(p^2-4)+3(p-2)+21$, i.e., $p=7$, which gives no solution.

Miscellaneous

1. John writes a number with 2187 digits on the blackboard, each digit being a 1 or 2. Judith creates a new number from John's number by reading the number from left to right and whenever she sees a 1 writing 112 and wherever she sees a 2 writing 111. (For example, if John's number begins 2112, then Judith's number would begin 111112112111.) After Judith finishes writing her number, she notices that the left most 2187 digits in her number and in John's number are the same. How many times do five 1's occur consecutively in John's number?

A. 182 B. 183 C. 184 D. 185 E. 186

1. Answer: A. Indeed John must start with a 1, and then everything follows. The 1 produce 112, then 112112111, then 112112111-112112111-112112112, notices the two 2s produces two 5 consecutive 1s. We have the table:

	Chains	Number of 1 produced	Number of 2 produced
3^1	112	2	1
3^2	112112111	7	2
3^3	112112111-112112111-112112112	20	7
3^4	----	40+21 = 61	20
3^5	----	122 + 60 = 182	61
3^6	----	364 + 183 = 547	182 (produces the chains of 5 consecutive 1s
$3^7 = 2187$			

2. A set of positive integers has the property that

Every member in the set, apart from 1, is divisible by at least one of 2, 3, or 5.

If the set contains $2n, 3n$ or $5n$ for some integer n, then it contains all three and n as well.

The set contains between 300 and 400 numbers. Exactly how many does it contain?

A. 324 B. 343 C. 351 D. 360 E. 364

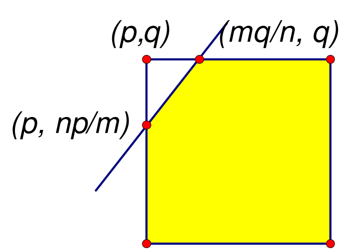
2. (E) Clearly the largest number is of the form 5^n. From 5^n we can generate all numbers of the form $2^r 3^s 5^t$ with $r+s+t=n$, possible choices are C_2^{n+2}, putting n identical balls into 3 boxes. From 5^n, can get 5^{n-1} and we can play the game again. Hence number in the set =

$$1+C_2^3+C_2^4+...+C_2^{n+2}=1+3+6+10+15+21+28+36+45+55+66+78=364.$$

3. Given a rectangular grid of m units wide and n units tall, where m and n are positive integers with $2n<m<3n$. The region below the line linking (0, 0) and (m, n) is shaded. There is a unit square whose shaded area is greater than 0.999 (but not totally shaded). The smallest possible value of mn satisfies

A. $496 \le mn \le 500$ B. $501 \le mn \le 505$ C. $506 \le mn \le 510$ D. $511 \le mn \le 515$ E. $516 \le mn \le 520$

3. (Cayley 2009) From $2n<m<3n$, get $\frac{1}{3}<\frac{n}{m}<\frac{1}{2}$, thus the slope of the line linking $(0,0)$ and (m,n) is between $\frac{1}{3}$ and $\frac{1}{2}$. Clearly to have un-shaded area less than 0.001, the un-shaded part must occupy the top left corner of coordinates: (p,q):



The area of the un-shaded part satisfies $0<\frac{1}{2}(\frac{m}{n}q-p)(q-\frac{n}{m}p)<0.001$, simplifies to get $0<500(mq-pn)^2<mn.$ Thus $mn>500.$ We try

$mq - pn = 1$ and $mn > 500$ to get two cases: $mn = 504$ or 510. The second case produces $n = 15, m = 34$ and hence $q = 4, p = 9$, hence $mn = 510$. The second case does not work. Answer: C.

4. How many odd coefficients are there in the expansion of $(x^2 - x + 1)^{2009}$?

4. Since we are interested only in the parity of the coefficients, we take modulo 2 on the coefficients throughout (so every coefficient is either 0 or 1). Let $f(n)$ be the number of odd coefficients in the expansion of $(x^2 - x + 1)^n$. To find $f(2009)$, we need a series of observations.

Observation 1. $f(2n) = f(n)$.

This follows from the fact that for pairwise distinct $a_1, a_2, \ldots, a_n$, we have

$$(x^{a_1} + x^{a_2} + \cdots + x^{a_n})^2 = x^{2a_1} + x^{2a_2} + \cdots + x^{2a_n} + 2\sum_{i \neq j} x^{a_i} x^{a_j} \equiv x^{2a_1} + x^{2a_2} + \cdots + x^{2a_n}.$$

Observation 2. $f(2^k) = 3$ for all non-negative integers k.

This follows from the fact that $(x^2 - x + 1)^2 \equiv (x^2 + x + 1)^2 = x^4 + 2x^3 + 3x^2 + 2x + 1 \equiv x^4 + x^2 + 1$, so we have $(x^2 - x + 1)^{2^k} \equiv x^{2^{k+1}} + x^{2^k} + 1$ for all non-negative integers k.

Observation 3. $f(2^k a + b) = f(a) f(b)$ for positive integers a, b where $b < 2^{k-1}$.

As remarked in Observation 2, we have $(x^2 - x + 1)^{2^k a} \equiv (y^2 + y + 1)^a$ where $y = x^{2^k}$. On the other hand, $(x^2 - x + 1)^b$ is of degree $2b < 2^k$. We thus get the desired conclusion.

Observation 4. $f(2^k - 1) = \dfrac{2^{k+2} - (-1)^k}{3}$ for all positive integers k.

Observe the coefficients of $(x^2 - x + 1)^{2^k - 1}$ for $k = 1, 2, 3, \ldots$, we see the pattern as follows: 111, 1101011, $(110)^2(111)(011)^2$ (this means $(x^2 - x + 1)^{2^3 - 1} \equiv (1x^0 + 1x^1 + 0x^2) + (1x^3 + 1x^4 + 0x^5)$ $+ (1x^6 + 1x^7 + 1x^8) + (0x^9 + 1x^{10} + 1x^{11}) + (0x^{12} + 1x^{13} + 1x^{14})$, same for below), $(110)^5(1)(011)^5$, $(110)^{10}(111)(011)^{10}$, …, and we can prove by induction that such pattern (i.e. $(110)^{\frac{2^k-2}{3}}(111)(011)^{\frac{2^k-2}{3}}$ for odd k and $(110)^{\frac{2^k-1}{3}}(1)(011)^{\frac{2^k-1}{3}}$ for even k) does indeed carry on.

With the above observations, we can proceed to find $f(2009)$. Since 2009 is 11111011001 in base 2, we have $2009 = 2^5(2^5 - 1) + 2^3(2^2 - 1) + 1$. It follows from the above observations that

$$f(2009) = f(2^5 - 1) f(2^2 - 1) f(1) = \frac{2^7 + 1}{3} \times \frac{2^4 - 1}{3} \times 3 = 645.$$

Remark. This is a difficult question. While an apparently similar question appeared in last year's Prelim (on binomial expansion), this question (on trinomial expansion) involves considerably more techniques. Considering the binary notation of 2009 should be the natural approach (especially if one works out the first few powers and finds that $f(2^k)=3$ for all k), and the most difficult part should be to find $f(2^k-1)$ in Observation 4, as it takes effort to compute enough data to look for patterns.

The general case of finding the number of odd coefficients of $(x^2+x+1)^n$ is a short-listed problem of IMO 1988.)

5. Let n be a positive integer and let a be an integer with $\gcd(a,n)=1$.
   Prove that

$$\frac{a^{\phi(n)}-1}{n} \equiv \sum_{\substack{j=1\\ \gcd(j,n)=1}}^{n} \frac{1}{aj}\left[\frac{aj}{n}\right] \pmod{n},$$

where $\phi(n)$ is the Euler's totient function and $[x]$ is the greatest integer not exceeding x.

5. Let $1 \le r_j \le n$ be the least nonnegative residue of aj modulo n for every $j \in \mathbf{Z}$.
Then the set

$$\{r_j : 1 \le j \le n,\ \gcd(j,n)=1\}$$

coincides with the set

$$\{j : 1 \le j \le n,\ \gcd(j,n)=1\},$$

since if $j_1 \not\equiv j_2 \pmod{n}$ then $r_{j_1} \ne r_{j_2}$.
Therefore

$$\begin{aligned}
a^{\phi(n)} &= \prod_{\substack{j=1\\ \gcd(j,n)=1}}^{n} \frac{aj}{j} \\
&= \prod_{\substack{j=1\\ \gcd(j,n)=1}}^{n} \frac{\left[\frac{aj}{n}\right]n + r_j}{j} \\
&= \prod_{\substack{j=1\\ \gcd(j,n)=1}}^{n} \frac{r_j}{j}\left(1+\frac{\left[\frac{aj}{n}\right]n}{r_j}\right) \\
&= \prod_{\substack{j=1\\ \gcd(j,n)=1}}^{n} \left(1+\frac{\left[\frac{aj}{n}\right]n}{r_j}\right) \\
&\equiv 1 + n\sum_{\substack{j=1\\ \gcd(j,n)=1}}^{n} \frac{1}{r_j}\left[\frac{aj}{n}\right] \pmod{n^2}
\end{aligned}$$

$$\equiv 1+n \sum_{\substack{j=1 \\ \gcd(j,n)=1}}^{n} \frac{1}{aj}\left[\frac{aj}{n}\right] \quad (\text{mod}\, n^2).$$

By subtracting 1 from both sides and dividing by n, one obtains the required congruence relation.

6. First note that $(-k)^2 \equiv (p-k)^2 \equiv k^2 \ (\text{mod}\, p)$. Also if $x^2 \equiv y^2 (\text{mod}\, p)$, where $1 \le x, y \le \frac{p-1}{2}$, then $(x-y)(x+y) \equiv 0 \ (\text{mod}\, p)$. But $1 < x+y < p$, hence $x = y$. These imply $1^2, 2^2, \cdots, \left(\frac{p-1}{2}\right)^2$ is a reduced residue system modulo p. Now since $p \equiv 1 (\text{mod}\, 4)$, by Euler's criterion, $\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}} = 1$, hence -1 is a square. Because $\left(\frac{-b}{p}\right) = \left(\frac{-1}{p}\right)\left(\frac{b}{p}\right)$, thus b is a square if and only if $-b \equiv p-b (\text{mod}\, p)$ is. Therefore the set $\{1^2, 2^2, \cdots, \left(\frac{p-1}{2}\right)^2\}$ consists of pairs $\{a_1, p-a_1, a_2, p-a_2, \cdots, a_{\frac{p-1}{4}}, p-a_{\frac{p-1}{4}}\}$ modulo p. As $1 < a_i, p-a_i < p$, we have $\{\frac{a_i}{p}\} + \{\frac{p-a_i}{p}\} = \frac{a_i + p - a_i}{p} = 1$, thus the sum is $\frac{p-1}{4}$.

Quadratic Congruence

1. $(a, p) = 1$ and p an odd prime of the form $4k+3$. Show that the solutions of $x^2 \equiv a \ (\text{mod}\, p)$ are $\pm a^{k+1}$.

1. Euler criterion $a^{2k+1} \equiv 1 \,(\text{mod}\, p)$ to get the answers.
2. a a quadratic residue of the odd prime p, show that $-a$ is a quadratic residue modulo p iff $p \equiv 1 \,(\text{mod}\, 4)$.
2. -1 a residue iff $p \equiv 1 \,(\text{mod}\, 4)$.

3. (a) p an odd prime, q the smallest positive quadratic nonresidue modulo p, then q is prime. (b) p an odd prime, q the smallest positive nonresidue modulo p, then $q < \sqrt{p} + 1$.

3. (a) If $q = ab$, then one of them is a nonresidue. (b) Let k be the smallest number such that $kq > p$, and $kq \equiv r \,(\text{mod}\, p)$, (remainder). As $(k-1)q < p$, have $r < q$, and hence r is

a residue and kq a residue. Therefore $k \geq q$, for $1 \leq k < q$, then k is a nonresidue and so kq is a nonresidue of p. Thus $(q-1)q < p$, it follows that $q < \sqrt{p}+1$.

4. There exist infinitely many primes of the form $4k+1$.

4. Consider $4(p_1p_2...p_N)^2+1$.

5. $83 \mid 2^{41}-1$?

5. Check $\left(\frac{2}{83}\right) = -1$, hence $2^{41} \equiv -1 \pmod{83}$ (Euler's criterion), i.e. $83 \mid 2^{41}+1$ but not $2^{41}-1$.

6. $1999 \mid 2^{999}-1$?

6. Yes, Euler's criterion.

7. q is odd and $p = 4q+1$ is prime. (i) 2 is a quadratic nonresidue of p. (ii) $p \mid 4^q+1$.

7. (i) $q = 2k+1$, then $p = 8k+5$ and 2 is a nonresidue. (ii) Euler: $-1 = \left(\frac{2}{p}\right) = 2^{\frac{p-1}{2}} = 2^{2q} = 4^q \pmod{p}$, hence the result.

8. Let p be an odd prime, find the number of quadratic nonresidue of p^n.

8. Residues are squares between 1 and p^n and are relatively prime to p. But if $x^2 \equiv a \pmod{p^n}$ is solvable, then it has 2 solutions, answer: $\frac{\phi(p^n)}{2} = \frac{p^{n-1}(p-1)}{2}$.

9. 3 is a nonresidues for all primes of the from 4^n+1.

9. Consider $\left(\frac{3}{p}\right)$. Note $p \equiv 2 \pmod{3}$.

10. p a prime of the form $8k+3$, $p \mid 2^{\frac{p-1}{2}}-1$?

10. No, Euler's criterion.

11. $1!+2!+...+n!$ is never a square if $n > 3$.

11. Note the expression $\equiv 3 \pmod{5}$, thus if $m^2 \equiv 3 \pmod{5}$, yet 3 is a nonresidue!

12. There exist infinitely many primes of the form (i) $8k+3$, (ii) $8k+5$ and (iii) $8k+7$.

12. (i) Consider $(p_1p_2...p_N)^2+2$, (ii) $(p_1p_2...p_N)^2+4$, (iii) $(p_1p_2...p_N)^2-2$.

Orders, Primitive Roots

4. 6 is a primitive root modulo 41, find the least positive residues of all elements of order 8 modulo 41.
1. All elements are of the form $6^k, 1 \leq k \leq 40$, and order of 6^k is $\frac{40}{(k,40)}$. It is of order 8 if $(k,40) = 5$, or $k = 5,15,25,35$. Then 6^k are 27,3,14 and 38.
2. If p is an odd prime, then the order of a modulo p is 2 iff $a \equiv -1 \pmod{p}$.
5. Clear, look for solutions of $x^2 \equiv 1 \pmod{p}$, 2 of them.

6. g a primitive root modulo p. If $p \equiv 3 \pmod 4$, then $-g$ is of order $\dfrac{p-1}{2}$.

3. g is not a residue, hence $g^{\frac{p-1}{2}} \equiv -1 \pmod p$. This implies $g^{\frac{p-1}{2}} g \equiv -g \pmod p$. But $p = 4k+3$, get $g^{2k+2} \equiv -g \pmod p$. But $(2k+2, 4k+2) = 2$, hence $-g$ is of order $\dfrac{p-1}{2}$.

4. $a_1, a_2, ..., a_r$ a reduced residue system modulo m, then $a_1^n, a_2^n, ..., a_r^n$ a reduced residue system modulo m iff $(n, \phi(m)) = 1$.

4. Easy.

5. Solve $x^5 \equiv 6 \pmod{101}$.

5. Calculate $2^{70} \equiv 6 \pmod{101}$, thus 2^{14} is a solution. As $(14,100) = 2$, we find 4 other solutions, 34, 54, 74, 94.

6. m is a positive integer having a primitive root, $(a, m) = 1$, then $x^k \equiv a \pmod m$ has a solution iff $a^{\frac{\phi(m)}{(k,\phi(m))}} \equiv 1 \pmod m$. Moreover if $x^k \equiv a \pmod m$ has a solution, then it has $(k, \phi(m))$ of them.

6. $d = (k, \phi(m))$ and g a primitive root. Taking indices, we see $x^k \equiv a \pmod m$ is solvable iff $k \operatorname{ind} x \equiv \operatorname{ind} a \pmod{\phi(m)}$. This is solvable for ind$x$ iff $d \mid \operatorname{ind} a$ and if solutions exist, there are d of them.

7. $p = 3k+2$, then a is a cubic residue.

7. a a cubic residue iff $a^{\frac{p-1}{(3,p-1)}} \equiv 1 \pmod p$. But $(3, p-1) = 1$. (Or $3\mu + (p-1)\nu = 1$, hence $a \equiv a^{3\mu+(p-1)\nu} \equiv (a^\mu)^3$.)

8. $p = 3k+1$, there are precisely one-third of the integers $1, 2, ..., p-1$ are cubic residues.

8. $\dfrac{\phi(p)}{(3,\phi(p))} = \dfrac{p-1}{(3, p-1)} = \dfrac{3k}{3} = k$ and one-third of them.

9. Find all primitive roots modulo 41.

9. 2 is one of them. The other primitive roots are 2^k, where $(k, 28) = 1$. We get 2, 8, 3, 19, 18, 14, 27, 21, 26, 10, 11, 15.

10. Which integer has exactly one primitive root?

10. We need $\phi(\phi(m)) = 1$. Now $\phi(k) = 1$ if $k = 1, 2$ and $\phi(m) = 1, 2$ if $m = 2, 3, 4, 6$. And -1 is the only primitive root.

11. g a primitive root of m, the quadratic residues are even powers of g and nonresidues odd powers of g.

12. If g^k a primitive root of m, then g is a primitive root modulo m.

13. -3 is not a primitive root of any prime of the form $4^n + 3$.

13. Check $\left(\dfrac{-3}{p}\right) = 1$.

14. Find all prime numbers p and q for which $a^{3pq} \equiv a \pmod{3pq}$ for all integers a.

14. (Rom 1996) We have $a^{3pq} \equiv a \pmod 3$, in particular $2^{3pq-1} \equiv 1 \pmod 3$, hence $2 \mid 3pq-1$, p and q are odd. Now $a^{3pq} \equiv a \pmod p$, if u is a primitive root modulo p, then $p-1 \mid 3pq-1$. Similarly $q-1 \mid 3pq-1$. Now $\frac{3pq-1}{p-1} = 3q + \frac{3q-1}{p-1}$, or $\frac{3q-1}{p-1}$ is an integer. If $p = q$, then $p = q = 3$. But $4^3 \equiv 1 \pmod 9$, or $4^9 \equiv 1 \pmod{27}$ and $4^{27} \equiv 1 \not\equiv 4 \pmod{27}$. Hence $p \neq q$. Now assume $q > p$, then $q \geq p+2 \Rightarrow \frac{3q-1}{p-1} < 3 \Rightarrow \frac{3q-1}{p-1} = 2 \Rightarrow q = \frac{3p+1}{2}$. As $\frac{3q-1}{p-1}$ is an integer, $\frac{9}{2} + \frac{10}{2(p-1)}$ is an integer, get $p-1 \mid 10$, or $p = 11, q = 17$. Check the conditions.

15. Let m be a positive integer. (a) Prove that if $2^{m+1}+1$ divides $3^{2^m}+1$, then $2^{m+1}+1$ is a prime. (b) Is the converse true?

15. (Kor 2003) Let $q = 2^{m+1}+1$, show that $\operatorname{ord}_q(3) = 2^{m-1} = q-1$. By Euler $\operatorname{ord}_q(3)$ is a divisor of $\phi(q) \leq q-1$, so must have $\phi(q) = q-1$, and q is prime. (b) Yes, $q = 2^{m+1}+1$ is prime, then $q \equiv 1 \pmod 4$ and $q \equiv 2 \pmod 3$, use quadratic reciprocity to show $\left(\frac{3}{q}\right) = (-1)^{\frac{(3-1)(q-1)}{4}} \left(\frac{q}{3}\right) = \left(\frac{2}{3}\right) = -1$, and $-1 = \left(\frac{3}{q}\right) \equiv 3^{\frac{q-1}{2}} \pmod q$ to get $q \mid 3^{2^m}+1$.

March2017